



How Well Do ACGME Milestones Track Readiness for Anesthesiology Certifying Examinations?

National cohort study of Anesthesiology Milestones 2.0 and ABA certification examination performance

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How Well Do Accreditation Council for Graduate Medical Education Milestones Track Readiness for Anesthesiology Certifying Examinations in a National Resident Cohort?

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Disclosures

- Ann Harman, PhD & Tianpeng Ye, PhD are current full-time employees of the ABA
- Stacie G. Deiner, MD is an ABA Director and former Chair of the ABA's Research Committee
- Huaping Sun, PhD, MBA is a former employee of the ABA
- No financial conflicts of interest



Our Mission

**To advance the highest standards
of the practice of anesthesiology**



Do Milestones Ratings Reflect Readiness for ABA Certification?

- **ACGME Milestones** track progression toward competence during residency training and readiness for independent practice
- **ABA certification examinations** assess whether a physician anesthesiology meets the highest national standards for the practice of anesthesiology
- **Study question:** Do these two assessment systems align?



Why This Question Matters

- Competency-based education (CBME) depends on meaningful assessment
- Program-based ratings guide resident feedback and progression decisions
- Certification examinations provide standardized national evidence
- Alignment helps clarify how Milestones data should be interpreted



Study Objective and Design

- Compared final Milestones 2.0 ratings with ABA examination performance
- National cohort of academic year 2021–2022 anesthesiology graduates
- Used final CA-3 ratings recorded before graduation
- Examined both overall and domain-specific associations



National Cohort of 2022 Residency Graduates

1,849

Residents with all
23 final Milestones ratings

1,799

Matched first-time
ADVANCED exam takers

1,383

Matched first-time
SOE and OSCE takers

First senior cohort evaluated under Anesthesiology Milestones 2.0

Assessments Compared

ACGME Milestones 2.0

- Program-based ratings by clinical competency committees
- 23 Milestones across 6 competencies

ABA Certification Examinations

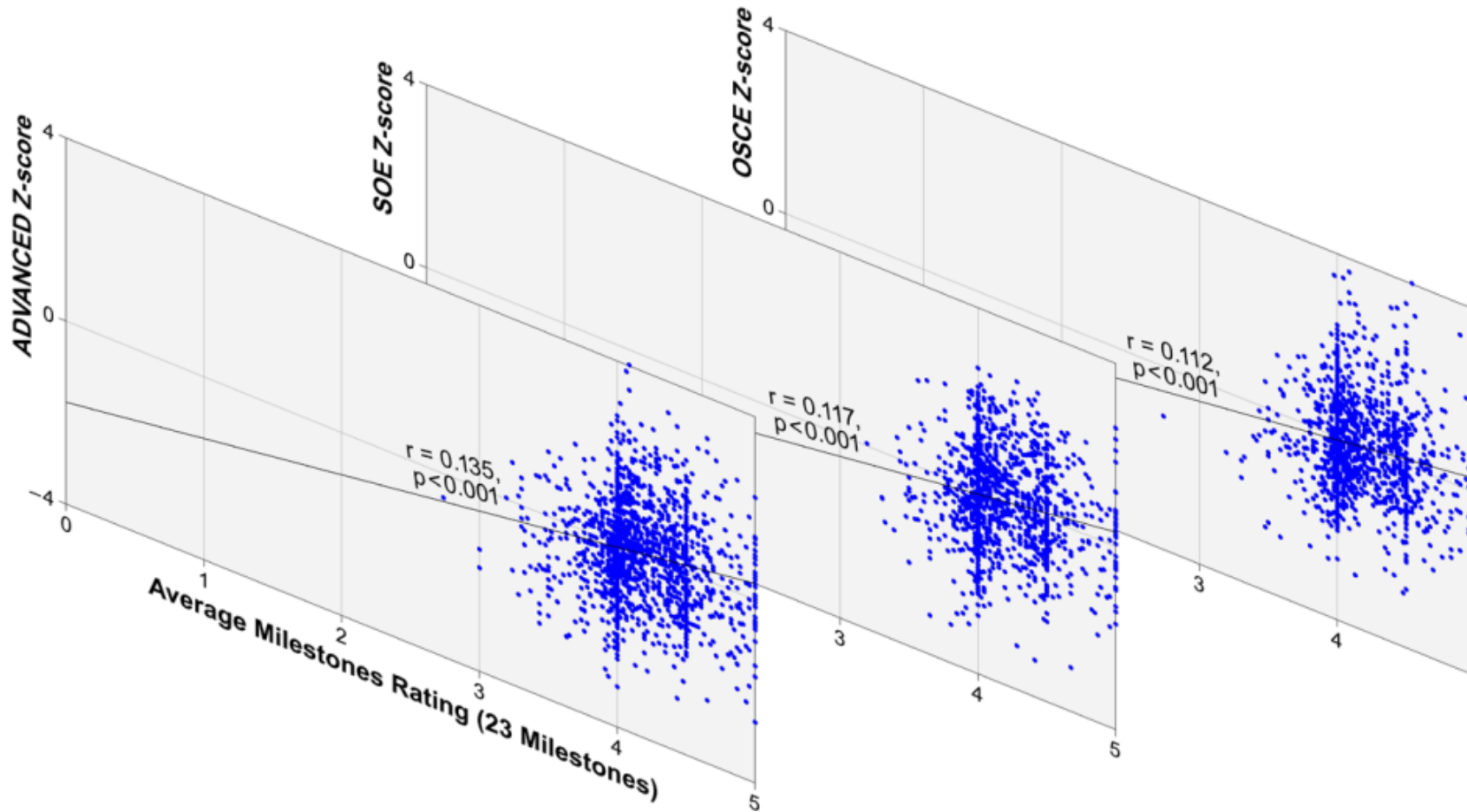
- ADVANCED Examination
- Standardized Oral Examination (SOE)
- Objective Structured Clinical Examination (OSCE)

How Alignment Was Tested

- **Overall:** Average of all 23 Milestones compared with examination performance
- **Domain-specific:** Selected Milestones compared with related examination domains
- Examination performance expressed as **standardized scores**
- **Correlations** used to evaluate strength of association



Overall Correlations



Overall Correlations: Statistically Significant but Weak

- ADVANCED: $r = 0.135$
- SOE: $r = 0.117$
- OSCE: $r = 0.112$
- Statistically significant, but limited practical predictive value
- Each explains under 2% of examination score variance

Conceptually Related Domain Alignment

Table 1. Conceptually Related Domains Evaluated by the ACGME Anesthesiology Milestones 2.0 and Tested by the ABA Certification Examinations

Competency area (no. of milestones)	Milestone description		Conceptually related domains tested by the ABA certification exams
Patient Care (10)	PC_Q1	Pre-Anesthetic Evaluation	SOE—Pre-Operative Evaluation
	PC_Q2	Peri-Operative Care and Management	
	PC_Q3	Application and Interpretation of Monitors	OSCE—Interpretation of Monitors (and Echocardiograms)
	PC_Q4	Intra-Operative Care	SOE—Intra-Operative Management
	PC_Q5	Airway Management	
	PC_Q6	Point-of-Care Ultrasound	OSCE—Application of Ultrasonography
	PC_Q7	Situational Awareness and Crisis Management	
	PC_Q8	Post-Operative Care	SOE—Post-Operative Care
	PC_Q9	Critical Care	
	PC_Q10	Regional (Peripheral and Neuraxial) Anesthesia	
Medical Knowledge (2)	MK_Q1	Foundational Knowledge	ADVANCED Examination (written)
	MK_Q2	Clinical Reasoning	ADVANCED Examination (written)
Systems-Based Practice (3)	SBP_Q1	Patient Safety and Quality Improvement	
	SBP_Q2	System Navigation for Patient-Centered Care	
	SBP_Q3	Physician Role in Health Care Systems	
Practice-Based Learning & Improvement (2)	PBLI_Q1	Evidence-Based and Informed Practice	
	PBLI_Q2	Reflective Practice and Commitment to Personal Growth	
Professionalism (3)	PR_Q1	Professional Behavior and Ethical Principles	OSCE—Ethical Issues
	PR_Q2	Accountability/Conscientiousness	
	PR_Q3	Well-Being	
Interpersonal & Communication Skills (3)	ICS_Q1	Patient- and Family-Centered Communication	
	ICS_Q2	Interprofessional and Team Communication	OSCE—Communication with other Professionals
	ICS_Q3	Communication within Health Care Systems	

Abbreviations: ABA, American Board of Anesthesiology; ACGME, Accreditation Council for Graduate Medical Education; OSCE, Objective Structured Clinical Examination; SOE, Standardized Oral Examination.

9 conceptually-related ACGME domains tested by the ABA certification exams

Conceptually Related Domain Correlations

Table 3. Correlations Between the ACGME Anesthesiology Milestone 2.0 Ratings and Related Domains Tested by the ABA Certification Examinations

Dyadic comparisons	Correlation	χ^2 test of independence	
		χ^2	P value
ADVANCED Examination (written)			
Average of MK_Q1 and MK_Q2 vs ADVANCED Z-score	0.289 ^{a***}	N/A	N/A
Structured Oral Examination (SOE)			
PC_Q1 vs Pre-Operative Evaluation	0.042 ^b	21.37	.810
PC_Q4 vs Intra-Operative Management	0.097 ^b	50.05	.551
PC_Q8 vs Post-Operative Care	0.027 ^b	35.44	.309
Objective Structured Clinical Examination (OSCE)			
PC_Q3 vs Interpretation of Monitors (and Echocardiograms)	0.093 ^b	22.67	.031*
PC_Q6 vs Application of Ultrasonography	0.052 ^b	8.12	.775
PR_Q1 vs Ethical Issues	0.049 ^b	39.00	.003**
ICS_Q2 vs Communication with other Professionals	0.052 ^b	26.65	.086

Abbreviations: ABA, American Board of Anesthesiology; ACGME, Accreditation Council for Graduate Medical Education; ICS, Interpersonal & Communication Skills; MK, Medical Knowledge; PC, Patient Care; PR, Professionalism.

* $P < .05$; ** $P < .01$; *** $P < .001$.

^aPearson correlation.

^bPolychoric correlation.

Why Might Alignment Be Limited?

- Milestones may capture typical clinical performance over time
- Examinations assess standardized, high-stakes performance
- Local implementation may vary across programs
- Narrow rating distributions may limit discrimination
- The two systems may provide related but distinct evidence



Implications for Competency-Based Assessment

- Interpret Milestones cautiously as indicators of ABA certification readiness
- Strengthen Milestones rating tools, faculty development, and CCC processes
- Improve feedback systems and assessment infrastructure
- Use program-based and certification evidence as complementary



